

Challenge 1 – Smart Salinity Ingress & Coastal Farmland Protection Advisor

Domain: Agriculture & Coastal Resilience

The Challenge

Coastal districts of Gujarat such as Bhavnagar, Jamnagar, and Kutch face increasing seawater ingress into groundwater and farmland, degrading soil quality and reducing crop yields. Farmers lack timely, localized guidance on salinity-resistant crops and mitigation practices.

Objective

Build Agentic AI solutions that monitor soil and groundwater salinity trends, recommend salttolerant crop alternatives, and guide farmers on land reclamation and irrigation practices.

Suggested AI Agents

- Soil & Groundwater Salinity Monitoring Agent
- Salinity-Resistant Crop Advisory Agent
- Irrigation & Land Reclamation Planning Agent
- Farmer Alert & Advisory Agent
- Coastal Farmland Health Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud